

Khizar Aamir

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EDUCATION

Bachelor of Science in Computer Science (Minor in Economics), Co-op Program Sep 2022 – May 2027
University of British Columbia Kelowna, BC

- **GPA: 4.2/4.33, Dean's List (2023–2025), Deputy Vice-Chancellor Scholarship (2023–2025)**
- **Relevant Coursework:** Web Programming, Software Engineering, Object Oriented Programming, Data Structures & Algorithms (Java), Data Analytics, Machine Architecture

EXPERIENCE

Software Engineer Intern Aug 2025 – Present
Aucctus AI Toronto, ON (Remote)

- Architected full-stack AI-powered features using **Django REST Framework** with PostgreSQL, **React/TypeScript**, and **Django Channels WebSockets** for real-time customer profile analysis and synthetic testing pipelines.
- Built scalable backend services with **Celery** task queues, **Docker** containerization, and **AWS** integration; implemented robust error handling and progress reporting for long-running AI workflows.
- Developed real-time WebSocket communication system with authentication, multitenancy support, and message validation.
- Implemented comprehensive testing suite with **PyTest** for backend APIs and **Vitest** for frontend components; established CI/CD pipelines with automated linting, type checking, and deployment workflows.

R&D Software Engineer Jan 2025 – Present
University of British Columbia Kelowna, BC

- Designed a **Generative AI simulation engine** using **GPT-4** for modeling student behavior; built prompt pipelines in **TypeScript/Python** with fine-tuned templates and **LLM-driven mastery adaptation**.
- Engineered sophisticated **student modeling algorithms** with personalized learning rates, dynamic difficulty adjustment using sigmoid functions, and topic-specific mastery calculations.
- Built production-ready **TypeScript/Node.js pipeline** with MySQL integration and automated CSV exports, processing **10K+** interaction logs with comprehensive error handling.

Full-Stack Engineer Intern Dec 2024 – May 2025
Strike Den Kelowna, BC (Remote)

- Engineered a **full-stack gym management platform** using **Next.js, MongoDB, Tailwind CSS** with **JWT authentication** and **RBAC**, streamlining class scheduling and trainer management workflows.
- Built a **responsive admin dashboard** enabling staff to manage trainers, classes, and weekly scheduling with **real-time updates** and optimistic UI patterns.
- Implemented **secure API endpoints** with **CSRF protection** and **rate limiting**, optimizing MongoDB queries with **schema indexing** to achieve **257ms median response times** under **50 concurrent users** with **99% reliability**.
- Built reusable React components with Tailwind CSS and Jest tests, ensuring responsive layouts and 100% unit-test coverage.

PROJECTS

ShakeScribe – AI-Powered Translator | Python, React, Node.js, Flask, Redis, DeepSeek LLM Dec 2024 – Mar 2025

- Built a **multi-model NLP pipeline** with sentiment analysis and semantic similarity matching engines, processing text through **custom linguistic scoring algorithms** for literary style assessment.
- Engineered **Shakespeare detection system** using **pattern recognition algorithms**, archaic vocabulary analysis, and **vector similarity calculations** to evaluate text authenticity and stylistic accuracy.
- Architected **Redis caching layer** with 7-day TTL and **API rate limiting** (10 requests/24hrs), orchestrating **microservices communication** between Node.js gateway and Python ML backend.

TSKFLO: Full-Stack Task Manager | React, Redux, Node.js, Express, Socket.io, MongoDB Jan 2025 – April 2025

- Built a **collaborative task management platform** featuring **JWT authentication**, **real-time Socket.io messaging**, and a **Dockerized REST API** with **15+ endpoints** for full CRUD operations.
- Implemented **role-based admin/user dashboards** with **automated task notifications**, reducing manual coordination by **70%** through streamlined workflows.
- Authored comprehensive API documentation and led peer code reviews, reducing onboarding time by **30%**.
- Iterated on Node.js/Express prototype features in two-week cycles, gathering feedback to refine endpoint designs.

TECHNICAL SKILLS

Languages: TypeScript/JavaScript (React, Node.js), Python, C# (fundamentals), SQL (PostgreSQL, MySQL, Java)
Frameworks & Tools: Next.js, .NET Core basics, Three.js (POC), REST, HTML/CSS
Testing & CI: Jest, PyTest, GitHub Actions / GitLab CI
Databases & Infra: MongoDB, PostgreSQL, Docker, Git
Practices: OOP design patterns, Agile/Scrum, code reviews, performance tuning